# 2.2 Complex Analysis: Problems

Attempt each problem for at least 30 minutes before reading solutions.md. Unless stated otherwise, circles are traversed once anticlockwise, $\mathbb{D}$ is the open unit disc and $\mathbb{H}$ the upper half-plane.

## Warm-up

**1.** (a) Show that $f(z) = \bar{z}$ is not complex differentiable at any point.

(b) Show that $g(z) = |z|^2$ is complex differentiable at $z = 0$ and nowhere else. Is $g$ holomorphic on any open set?

**2.** Find all holomorphic functions $f = u + iv$ on $\mathbb{C}$ with $u(x, y) = x^2 - y^2 + x$.

**3.** Find the radius of convergence of:

(a) $\sum_{n \ge 0} n^3 z^n$; (b) $\sum_{n \ge 1} z^{n!}$; (c) $\sum_{n \ge 0} \dfrac{(n!)^2}{(2n)!} z^n$.

**4.** Let $C$ be the unit circle. Compute $\int_C \bar{z}\, dz$ and $\int_C z^n\, dz$ for every integer $n$. Why does the first answer not contradict Cauchy's theorem?

**5.** Find the Laurent series of $f(z) = \dfrac{1}{z(z-1)}$ on (a) $0 < |z| < 1$ and (b) $|z| > 1$. What is $\operatorname{Res}(f, 0)$?

**6.** Show that $\varphi(z) = \dfrac{z - i}{z + i}$ is a conformal bijection from $\mathbb{H}$ onto $\mathbb{D}$, and find its inverse.

## Core

**7.** Let $f$ be entire and suppose there are constants $A, B$ and an integer $n \ge 0$ with $|f(z)| \le A + B|z|^n$ for all $z$. Show that $f$ is a polynomial of degree at most $n$.

**8.** Let $f$ be entire with $\operatorname{Re} f(z) \le M$ for all $z$. Show that $f$ is constant.

**9.** Show that $\displaystyle\int_{-\infty}^{\infty} \frac{dx}{1 + x^4} = \frac{\pi}{\sqrt{2}}$.

**10.** For $a > 1$, show that $\displaystyle\int_0^{2\pi} \frac{d\theta}{a + \cos\theta} = \frac{2\pi}{\sqrt{a^2 - 1}}$.

**11.** Show that $\displaystyle\int_{-\infty}^{\infty} \frac{\cos x}{x^2 + 1}\, dx = \frac{\pi}{e}$.

**12.** Show that $\displaystyle\int_0^{\infty} \frac{\sin x}{x}\, dx = \frac{\pi}{2}$ (as an improper Riemann integral, $\lim_{R \to \infty} \int_0^R$).

**13.** Classify the singularity at the given point and find the residue:

(a) $\dfrac{1}{z^2 \sin z}$ at $z = 0$;
(b) $e^{1/z}$ at $z = 0$;
(c) $\cot z$ at $z = k\pi$, $k \in \mathbb{Z}$.

**14.** Show that all five zeros of $p(z) = z^5 + 3z^3 + 7$ lie in the annulus $1 < |z| < 2$.

**15.** Let $\lambda > 1$. Show that the equation $ze^{\lambda - z} = 1$ has exactly one solution in $\mathbb{D}$, and that this solution is real and positive.

**16.** Show that every conformal bijection $f : \mathbb{D} \to \mathbb{D}$ with $f(0) = 0$ is a rotation $f(z) = e^{i\theta} z$.

**17.** Let $f$ be holomorphic on $\mathbb{D}$ and continuous on $\overline{\mathbb{D}}$, with $|f(z)| = 1$ for $|z| = 1$ and $f(z) \neq 0$ for $z \in \mathbb{D}$. Show that $f$ is constant. Show that the hypothesis $f \neq 0$ cannot be dropped.

**18.** (a) Is there a holomorphic function $f$ on $\mathbb{D}$ with $f(1/n) = (-1)^n/n$ for all $n \geq 2$?

(b) Find all holomorphic $f$ on $\mathbb{D}$ with $f(1/n) = 1/n^2$ for all $n \geq 2$.

**19.** Let $f$ be holomorphic on an open set containing $\overline{\mathbb{D}}$, with no zeros on $|z| = 1$. Let $a_1, \ldots, a_k$ be the zeros of $f$ in $\mathbb{D}$, listed with multiplicity. Show that

$$\frac{1}{2\pi i} \int_{|z|=1} \frac{z f'(z)}{f(z)}\, dz = a_1 + \cdots + a_k.$$

## Challenge

**20.** Show that every injective entire function has the form $f(z) = az + b$ with $a \neq 0$.

**21.** For $0 < a < 1$, show that

$$\int_0^\infty \frac{x^{a-1}}{1+x}\, dx = \frac{\pi}{\sin(\pi a)}.$$

**22.** For $|a| < 1$ let $\varphi_a(z) = \dfrac{z - a}{1 - \bar{a} z}$.

(a) Show that $\varphi_a$ maps $\mathbb{D}$ bijectively onto $\mathbb{D}$, with inverse $\varphi_{-a}$.

(b) Show that every conformal bijection $\mathbb{D} \to \mathbb{D}$ has the form $e^{i\theta}\varphi_a$ for some $\theta \in \mathbb{R}$, $|a| < 1$.

**23.** Let $f_n$ be holomorphic on an open set $\Omega$ and suppose $f_n \to f$ uniformly on every compact subset of $\Omega$. Show that $f$ is holomorphic on $\Omega$ and that $f_n' \to f'$ uniformly on every compact subset of $\Omega$.